```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.pipeline import make_pipeline
from sklearn.svm import SVR
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

df = pd.read_csv("/content/advertising.csv")

print(df.head())
print(df.tail())
print(df.shape)
print(df.dtypes)
print(df.isnull().sum())
print(df.duplicated().sum())
print(df.describe())

X = df[["TV", "Radio", "Newspaper"]]
y = df["Sales"]

sns.pairplot(df[["TV", "Radio", "Newspaper", "Sales"]])
plt.suptitle("Pair Plot of Advertising Features and Sales", y=1.02)
plt.show()

corr = df[["TV", "Radio", "Newspaper", "Sales"]].corr()

plt.figure(figsize=(7, 5))
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.5)
plt.title("Correlation Heatmap")
plt.show()
```

```
corr_sales = corr["Sales"].drop("Sales").abs().sort_values(ascending=False)
strongest = corr_sales.index[0]

print(corr["Sales"].drop("Sales"))
print(strongest)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

linear_model = LinearRegression()
linear_model.fit(X_train, y_train)
y_pred_linear = linear_model.predict(X_test)

poly_model = make_pipeline(
    PolynomialFeatures(degree=2),
    LinearRegression()
)
poly_model.fit(X_train, y_train)
y_pred_poly = poly_model.predict(X_test)

svr_model = make_pipeline(
    StandardScaler(),
    SVR(kernel="rbf")
)
svr_model.fit(X_train, y_train)
y_pred_svr = svr_model.predict(X_test)

predictions = {
    "Linear Regression": y_pred_linear,
    "Polynomial Regression": y_pred_poly,
    "SVR": y_pred_svr
}

results = []

for name, pred in predictions.items():
    mae = mean_absolute_error(y_test, pred)
```

```
    mae = mean_absolute_error(y_test, pred)
    mse = mean_squared_error(y_test, pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_test, pred)
    results.append([name, mae, mse, rmse, r2])

results_df = pd.DataFrame(
    results,
    columns=["Model", "MAE", "MSE", "RMSE", "R2 Score"]
)

print(results_df)

results_df.set_index("Model")[["MAE", "MSE", "RMSE", "R2 Score"]].plot(
    kind="bar", figsize=(10, 6)
)

plt.title("Model Performance Comparison")
plt.xlabel("Models")
plt.ylabel("Metric Value")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

best_model = results_df.loc[results_df["RMSE"].idxmin(), "Model"]

if best_model == "Linear Regression":
    best_predictions = y_pred_linear
elif best_model == "Polynomial Regression":
    best_predictions = y_pred_poly
else:
    best_predictions = y_pred_svr

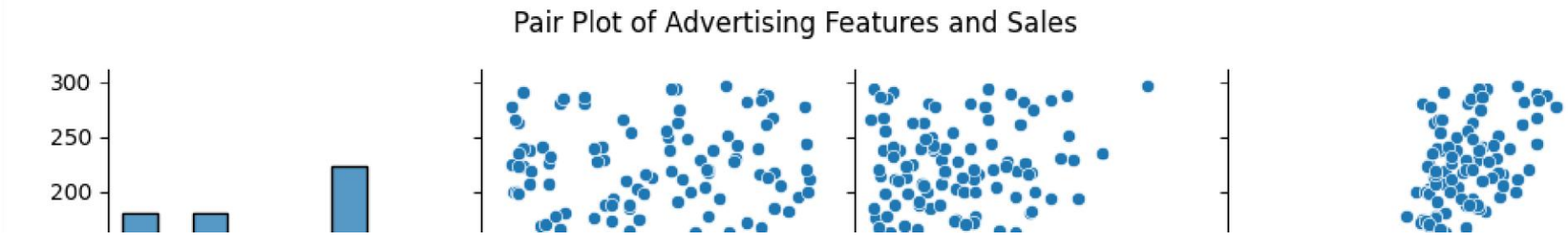
plt.figure(figsize=(8, 6))
plt.scatter(y_test, best_predictions)
plt.plot(
    [y_test.min(), y_test.max()],
    [y_test.min(), y_test.max()],
    linestyle='--'
```

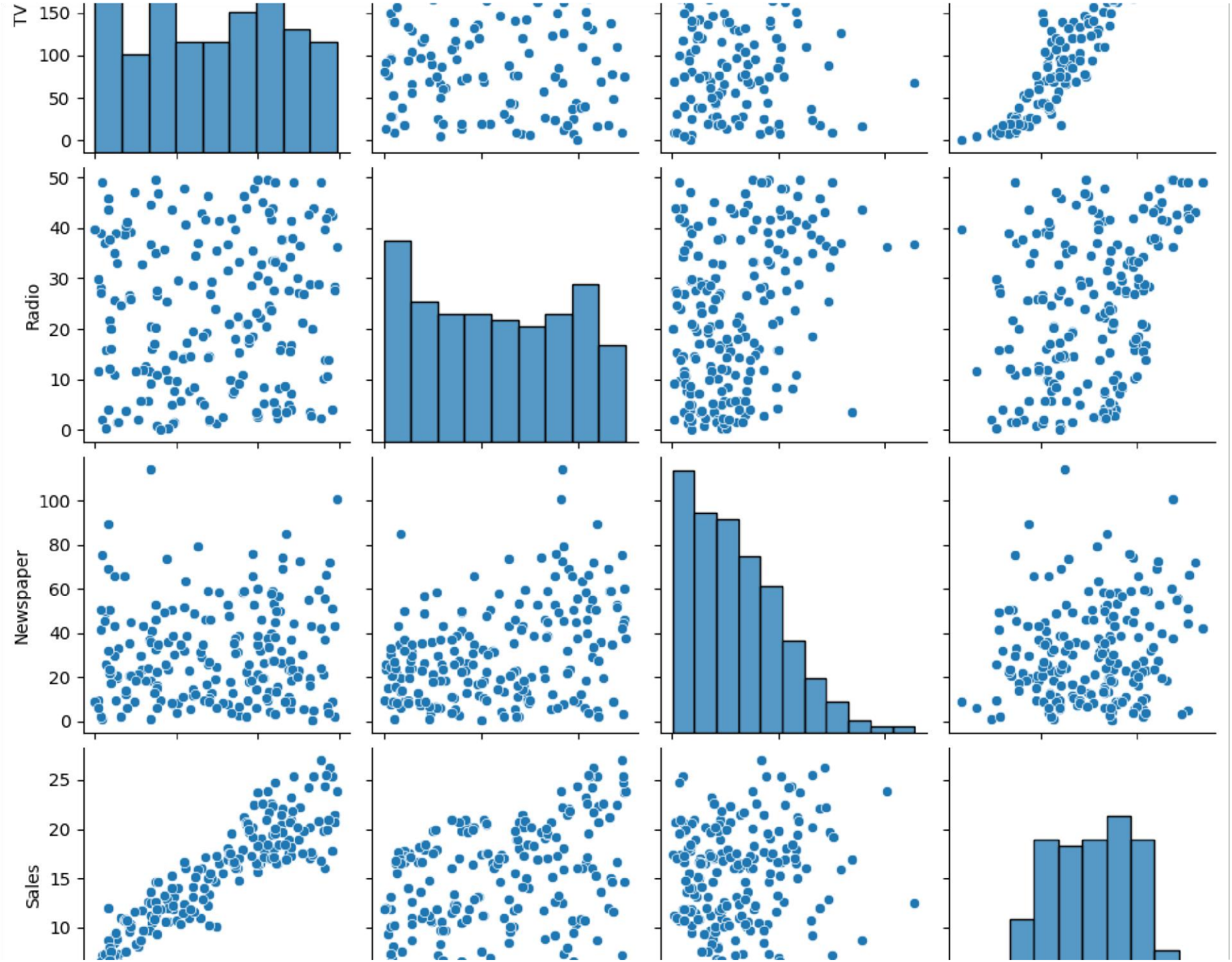
```
    linestyle= --
)
plt.xlabel("Actual Sales")
plt.ylabel("Predicted Sales")
plt.title("Actual vs Predicted Sales - " + best_model)
plt.tight_layout()
plt.show()

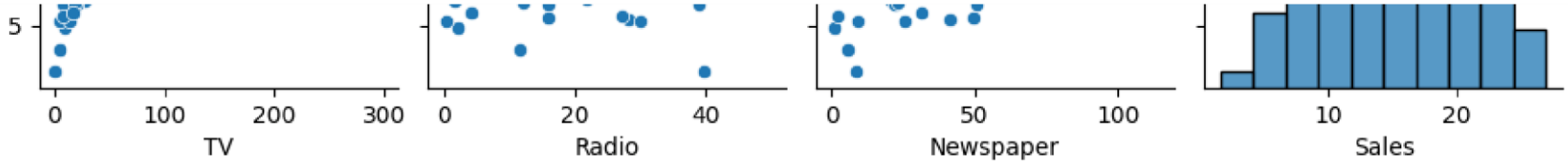
print("Best Performing Model:", best_model)

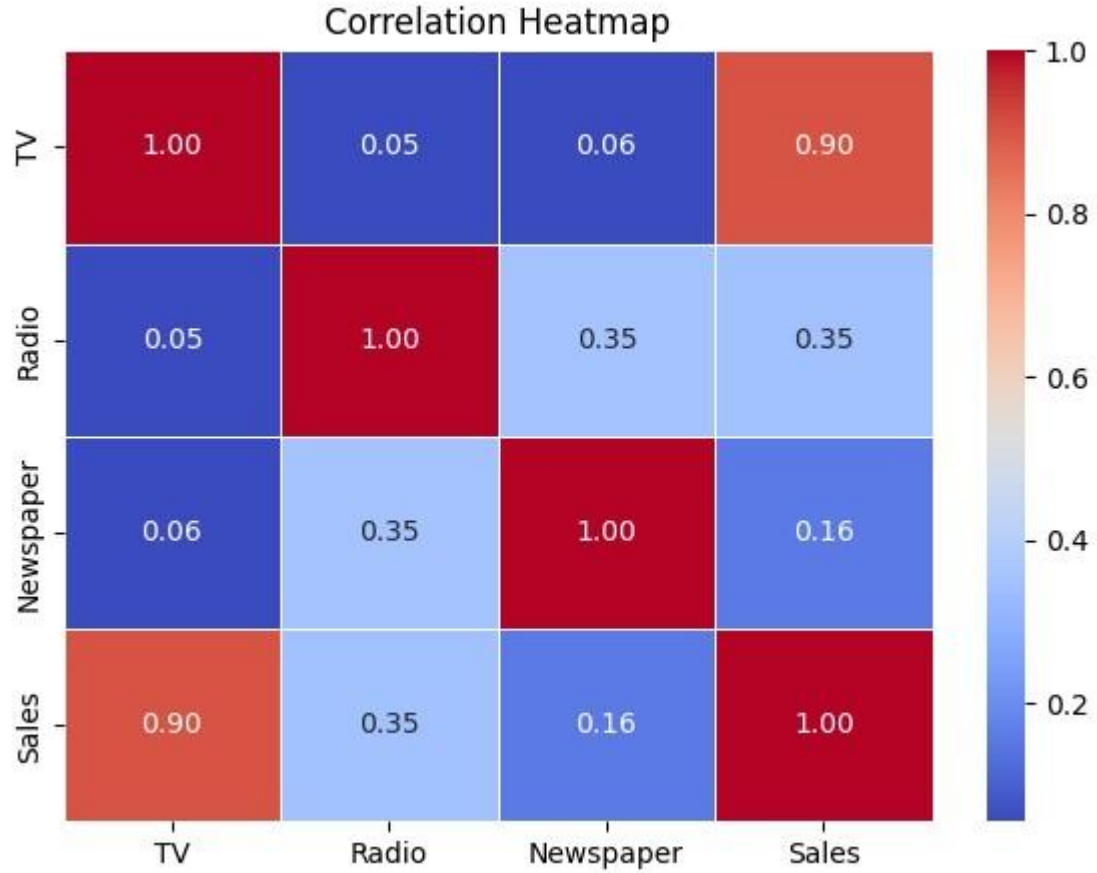
print("1. Allocate more budget to TV advertising.")
print("2. Use the best-performing model for Sales prediction.")
print("3. Focus on advertising channels with stronger impact on Sales.")
```

```
      TV  Radio  Newspaper  Sales
0  230.1   37.8       69.2   22.1
1   44.5   39.3       45.1   10.4
2   17.2   45.9       69.3   12.0
3  151.5   41.3       58.5   16.5
4  180.8   10.8       58.4   17.9
        TV  Radio  Newspaper  Sales
195   38.2    3.7       13.8    7.6
196   94.2    4.9        8.1   14.0
197  177.0    9.3        6.4   14.8
198  283.6   42.0       66.2   25.5
199  232.1    8.6        8.7   18.4
(200, 4)
TV           float64
Radio        float64
Newspaper    float64
Sales        float64
dtype: object
TV           0
Radio        0
Newspaper    0
Sales        0
dtype: int64
0
                TV       Radio   Newspaper       Sales
count  200.000000  200.000000  200.000000  200.000000
mean   147.042500   23.264000   30.554000   15.130500
std     85.854236   14.846809   21.778621    5.283892
min      0.700000    0.000000    0.300000    1.600000
25%     74.375000    9.975000   12.750000   11.000000
50%    149.750000   22.900000   25.750000   16.000000
75%    218.825000   36.525000   45.100000   19.050000
max    296.400000   49.600000  114.000000   27.000000
```



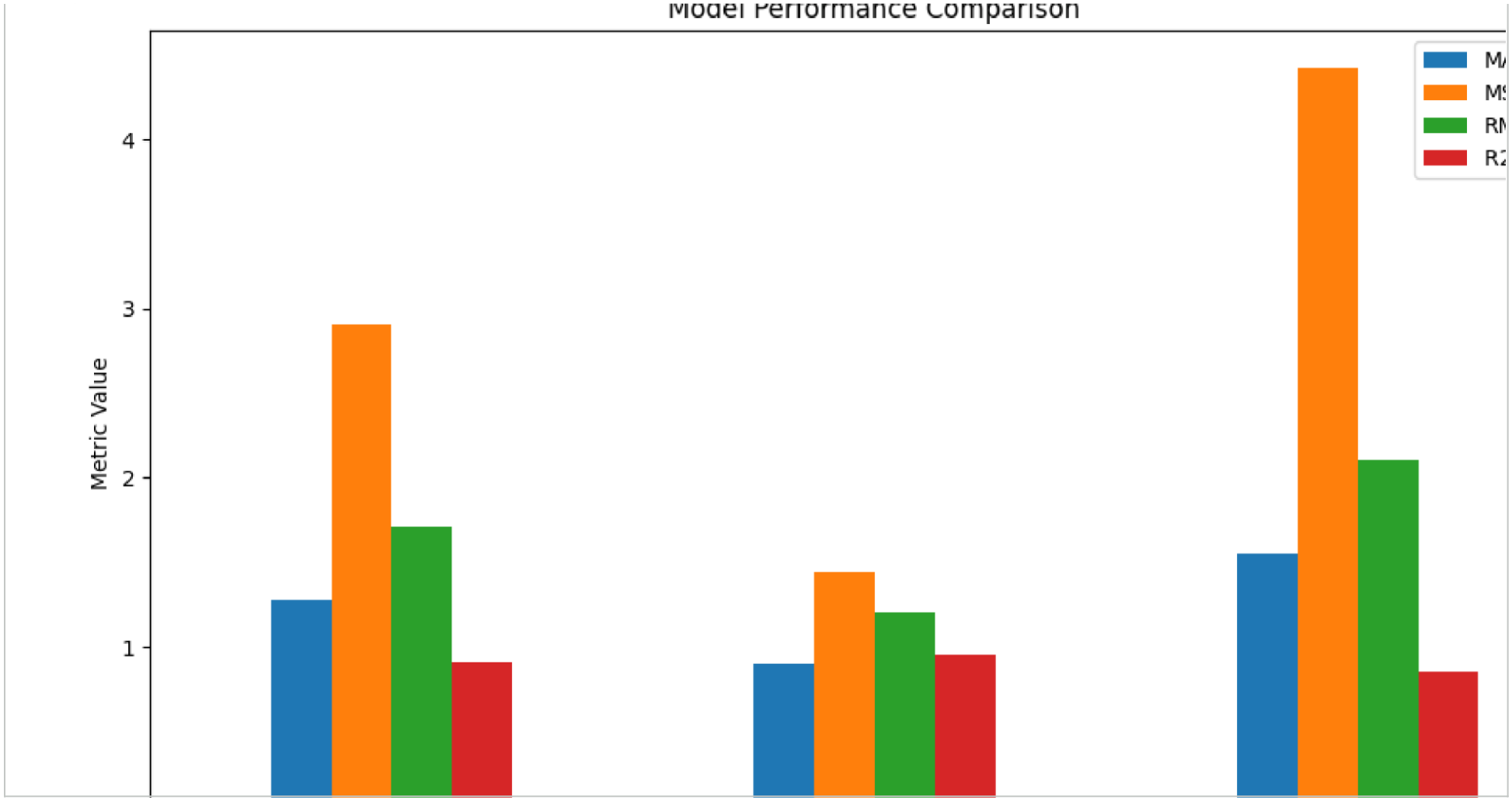

TV
Radio
Newspaper
Sales
0
50
100
150
0
10
20
30
40
50
0
20
40
60
80
100
10
15
20
25

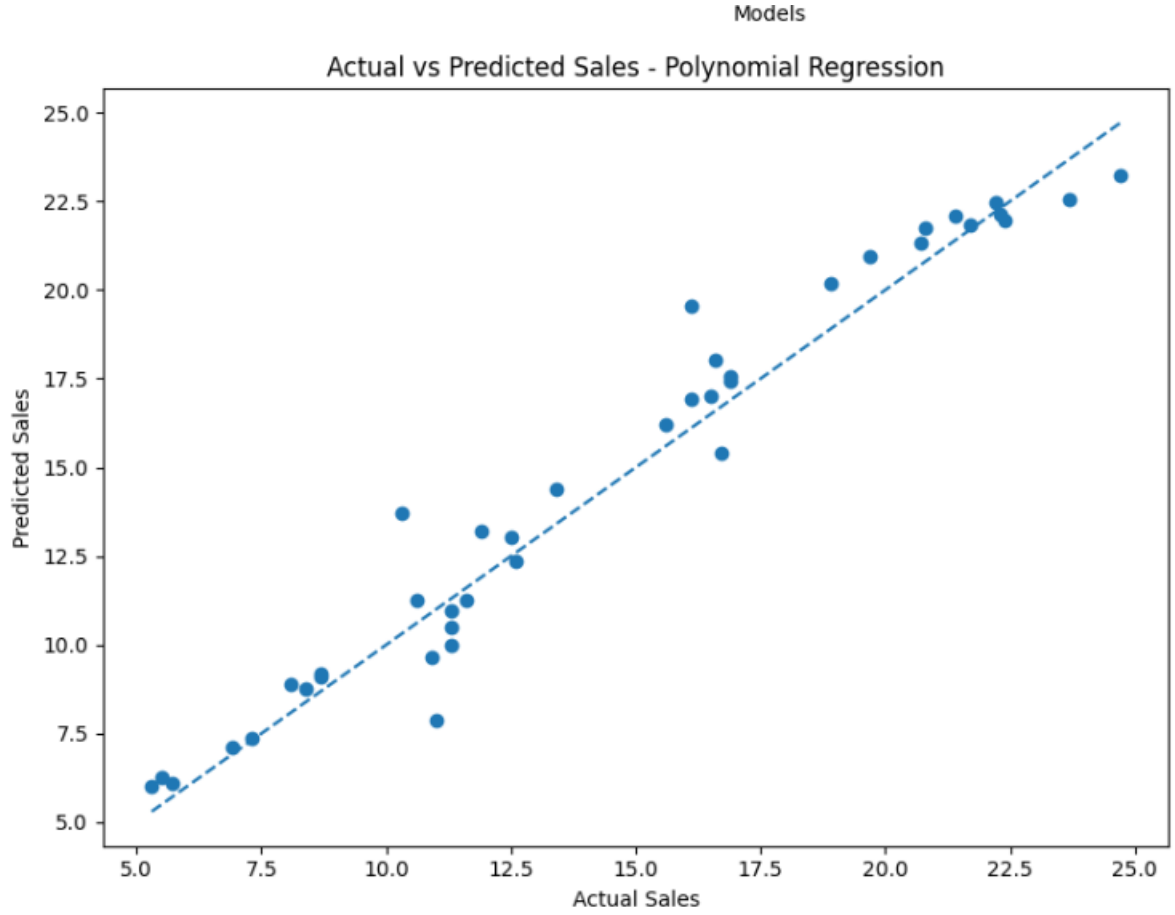





```
TV           0.901208
Radio        0.349631
Newspaper    0.157960
Name: Sales, dtype: float64
TV
```

	Model	MAE	MSE	RMSE	R2 Score
0	Linear Regression	1.274826	2.907757	1.705215	0.905901
1	Polynomial Regression	0.903371	1.442543	1.201059	0.953317
2	SVR	1.555355	4.421034	2.102626	0.856930


Model Performance Comparison
Metric Value
4
3
2
1




```
Best Performing Model: Polynomial Regression
1. Allocate more budget to TV advertising.
2. Use the best-performing model for Sales prediction.
3. Focus on advertising channels with stronger impact on Sales.
```